Laporan Praktikum 2. TCP/IP Model

Kegiatan : Identifikasi OSI Layer DNS www.pens.ac.id

Nama : Ramdhan Nur Afandi

NRP : 5325600019

Data Layer

Application layer

```
Domain Name System (query)
    Transaction ID: 0x0007
    Flags: 0x0100 Standard query
        0... .... .... .... = Response: Message is a query
        .000 0... .... .... = Opcode: Standard query (0)
        .... ..0. .... .... = Truncated: Message is not truncated
        .... ...1 .... .... = Recursion desired: Do query recursively
        .... .... .0.. .... = Z: reserved (0)
        .... .... ...0 .... = Non-authenticated data: Unacceptable
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    Queries
        www.pens.ac.id: type AAAA, class IN
            Name: www.pens.ac.id
            [Name Length: 14]
            [Label Count: 4]
            Type: AAAA (28) (IP6 Address)
            Class: IN (0x0001)
    [Response In: 561847]
```

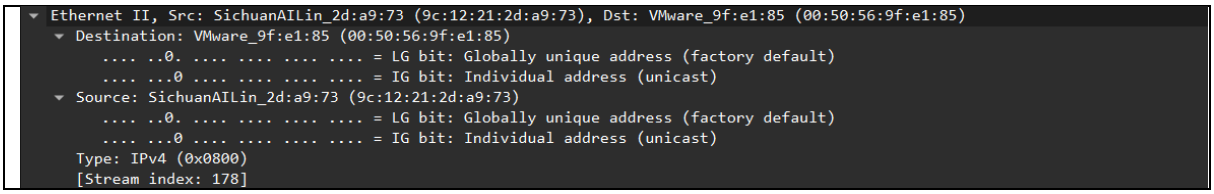
Host-to-host layer

```
User Datagram Protocol, Src Port: 61975, Dst Port: 53
    Source Port: 61975
    Destination Port: 53
    Length: 40
    Checksum: 0xd024 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 9645]
    [Stream Packet Number: 1]
    [Timestamps]
        [Time since first frame: 0.000000000 seconds]
        [Time since previous frame: 0.000000000 seconds]
    UDP payload (32 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 10.252.134.112, Dst: 10.252.128.3
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 60
    Identification: 0xa482 (42114)
    000. .... = Flags: 0x0
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x79c3 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.252.134.112
    Destination Address: 10.252.128.3
    [Stream index: 346]
```

Network Interface layer



Laporan Praktikum 2. TCP/IP Model

Kegiatan : Identifikasi OSI Layer FTP newfs.pens.ac.id

Nama : Ramdhan Nur Afandi

NRP : 5325600019

Data Layer

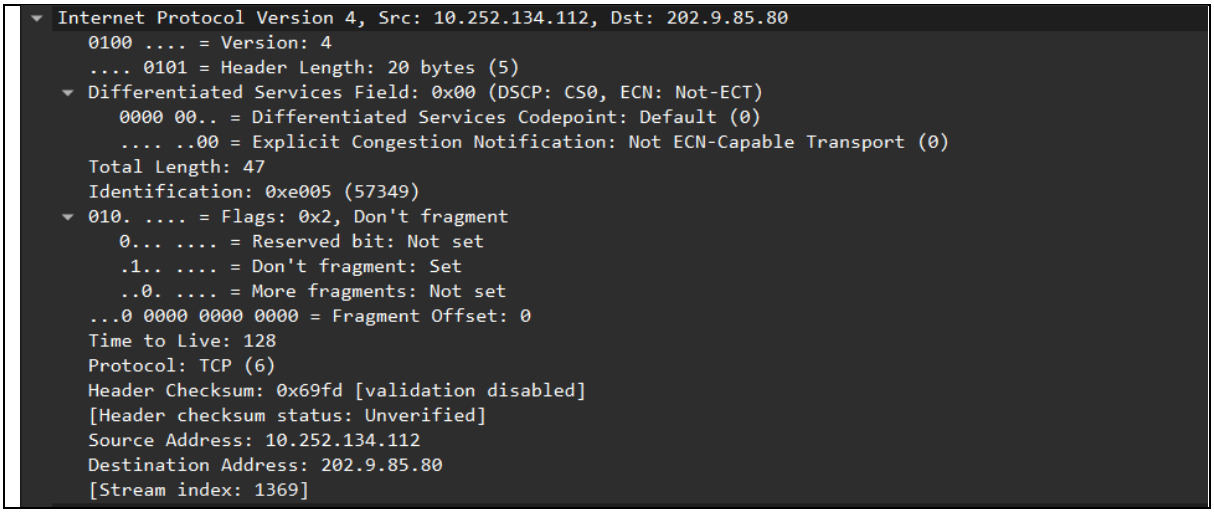
Application layer

```
File Transfer Protocol (FTP)
  PASS \r\n
      Request command: PASS
  [Current working directory: ]
```

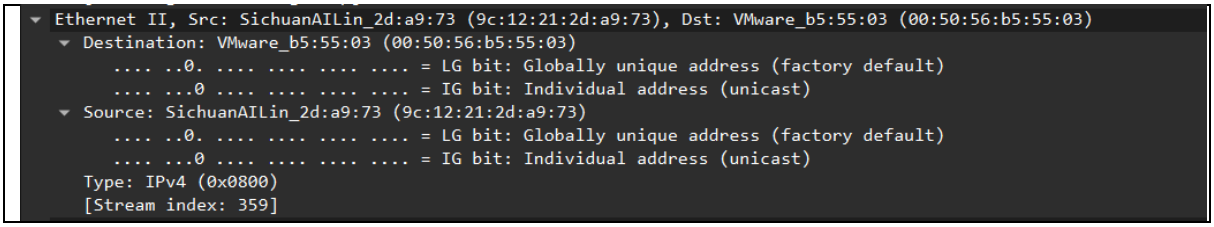
Host-to-host layer

```
Transmission Control Protocol, Src Port: 54313, Dst Port: 21, Seq: 31, Ack: 166, Len: 7
    Source Port: 54313
    Destination Port: 21
    [Stream index: 230]
    [Stream Packet Number: 12]
  [Conversation completeness: Complete, WITH_DATA (31)]
      ..0. .... = RST: Absent
      ...1 .... = FIN: Present
      .... 1... = Data: Present
      .... .1.. = ACK: Present
      .... ..1. = SYN-ACK: Present
      .... ...1 = SYN: Present
      [Completeness Flags: ·FDASS]
    [TCP Segment Len: 7]
    Sequence Number: 31    (relative sequence number)
    Sequence Number (raw): 2421685028
    [Next Sequence Number: 38    (relative sequence number)]
    Acknowledgment Number: 166    (relative ack number)
    Acknowledgment number (raw): 958060541
    0101 .... = Header Length: 20 bytes (5)
  Flags: 0x018 (PSH, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 1... = Push: Set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·····AP···]
    Window: 8027
    [Calculated window size: 8027]
    [Window size scaling factor: 1]
    Checksum: 0xa12f [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  [Timestamps]
      [Time since first frame in this TCP stream: 4.164133700 seconds]
      [Time since previous frame in this TCP stream: 487.140800 milliseconds]
  [SEQ/ACK analysis]
      [iRTT: 5.555900 milliseconds]
      [Bytes in flight: 7]
      [Bytes sent since last PSH flag: 7]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (7 bytes)
```

Internetwork layer



Network Interface layer



Laporan Praktikum 2. TCP/IP Model

Kegiatan : Identifikasi OSI Layer HTTP http://www.pens.ac.id

Nama : Ramdhan Nur Afandi

NRP : 5325600019

Data Layer

Application layer

```
▾ Hypertext Transfer Protocol
  ▾ GET / HTTP/1.1\r\n
      Request Method: GET
      Request URI: /
      Request Version: HTTP/1.1
    Host: www.pens.ac.id\r\n
    User-Agent: curl/8.19.0\r\n
    Accept: */*\r\n
    \r\n
    [Response in frame: 989552]
    [Full request URI: http://www.pens.ac.id/]
```

Host-to-host layer

```
▾ Transmission Control Protocol, Src Port: 60369, Dst Port: 80, Seq: 1, Ack: 1, Len: 78
    Source Port: 60369
    Destination Port: 80
    [Stream index: 299]
    [Stream Packet Number: 4]
  ▾ [Conversation completeness: Complete, WITH_DATA (31)]
      ..0. .... = RST: Absent
      ...1 .... = FIN: Present
      .... 1... = Data: Present
      .... .1.. = ACK: Present
      .... ..1. = SYN-ACK: Present
      .... ...1 = SYN: Present
      [Completeness Flags: ·FDASS]
    [TCP Segment Len: 78]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2078985138
    [Next Sequence Number: 79    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2347457539
    0101 .... = Header Length: 20 bytes (5)
  ▾ Flags: 0x018 (PSH, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 1... = Push: Set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·····AP···]
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0xff34 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
```

```
▾ [Timestamps]
      [Time since first frame in this TCP stream: 42.758500 milliseconds]
      [Time since previous frame in this TCP stream: 18.344300 milliseconds]
▾ [SEQ/ACK analysis]
      [iRTT: 24.414200 milliseconds]
      [Bytes in flight: 78]
      [Bytes sent since last PSH flag: 78]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (78 bytes)
```

Internetwork layer

```
▾ Internet Protocol Version 4, Src: 10.252.134.112, Dst: 202.9.85.176
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▾ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 118
    Identification: 0x1bac (7084)
  ▾ 010. .... = Flags: 0x2, Don't fragment
      0... .... = Reserved bit: Not set
      .1.. .... = Don't fragment: Set
      ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x2db0 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.252.134.112
    Destination Address: 202.9.85.176
    [Stream index: 3070]
```

Network Interface layer

```
▾ Ethernet II, Src: SichuanAILin_2d:a9:73 (9c:12:21:2d:a9:73), Dst: VMware_b5:55:03 (00:50:56:b5:55:03)
  ▾ Destination: VMware_b5:55:03 (00:50:56:b5:55:03)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  ▾ Source: SichuanAILin_2d:a9:73 (9c:12:21:2d:a9:73)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 359]
```